```
clc;
clear;
close all;

%% FIR FILTER PARAMETERS
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;

%% 1. DIRECT REALIZATION
yd = filter(b,1,x);

%% 2. CASCADE REALIZATION
z = roots(b);
g = b(1);
yc = g*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);

%% 3. SYMMETRY AND LINEAR PHASE
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

%% 4. FREQUENCY, PHASE AND GROUP DELAY
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);

%% 5. NUMERICAL COMPARISON
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;

%% 6. DISPLAY RESULTS
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf(' FIR FILTER REALIZATION RESULTS\n');
```

```
 FIR FILTER REALIZATION RESULTS
```

```
fprintf('=========================================\n');
```

```
=========================================
```

```
fprintf('\nFIR Coefficients:\n');
```

```
FIR Coefficients:
```

```
disp(b);
```

```
    0.0625    0.2500    0.3750    0.2500    0.0625
```

```
fprintf('Sampling Frequency = %d Hz\n',Fs);
```

```
Sampling Frequency = 1000 Hz
```

```
fprintf('Number of Samples = %d\n',N);
```

```
Number of Samples = 128
```

```
fprintf('Coefficient Symmetry = %s\n',symmetry);
```

```
Coefficient Symmetry = Symmetric
```

```
fprintf('Symmetry Error = %.3e\n',symErr);
```

```
Symmetry Error = 0.000e+00
```

```
fprintf('Group Delay = %.2f samples\n',D);
```

```
Group Delay = 2.00 samples
```

```
fprintf('\nMaximum Error = %.3e\n',err);
```

```
Maximum Error = 8.327e-16
```

```
fprintf('Percentage Deviation = %.3e %%\n',dev);
```

```
Percentage Deviation = 2.220e-13 %
```

```
%% 7. FIRST 10 IMPULSE RESPONSE VALUES
fprintf('\n=========================================\n');
```

```
=========================================
```

```
fprintf('FIRST 10 IMPULSE RESPONSE VALUES\n');
```

```
FIRST 10 IMPULSE RESPONSE VALUES
```

```
fprintf('=========================================\n');
```

```
==========================================
```

```
fprintf(' n      Direct      Cascade\n');
```

```
 n      Direct      Cascade
```

```
for k = 1:10
    fprintf('%2d    %10.6f  %10.6f\n', ...
        n(k),yd(k),yc(k));
end
```

```
 0      0.062500    0.062500
 1      0.250000    0.250000
 2      0.375000    0.375000
 3      0.250000    0.250000
 4      0.062500    0.062500
 5      0.000000    0.000000
 6      0.000000    0.000000
 7      0.000000    0.000000
 8      0.000000    0.000000
 9      0.000000    0.000000
```

```
%% 8. CASCADE ZEROS
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf('CASCADE REALIZATION ZEROS\n');
```

```
CASCADE REALIZATION ZEROS
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
disp(z);
```

```
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

```
%% 9. OUTPUT PARAMETER TABLE
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
};

Observation = {
    'Direct FIR filtering'
```

```
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
    'Equivalent transfer-function behavior'
};

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('Group delay = %.2f samples',D)
    sprintf('Max error = %.3e; Deviation = %.3e %%',err,dev)
};

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
};

Output_Table = table(Parameter,Observation,Value,Unit_Note);

fprintf('\n===========================================\n');
```

```
==========================================
```

```
fprintf('OUTPUT PARAMETER TABLE\n');
```

```
OUTPUT PARAMETER TABLE
```

```
fprintf('===========================================\n');
```

```
==========================================
```

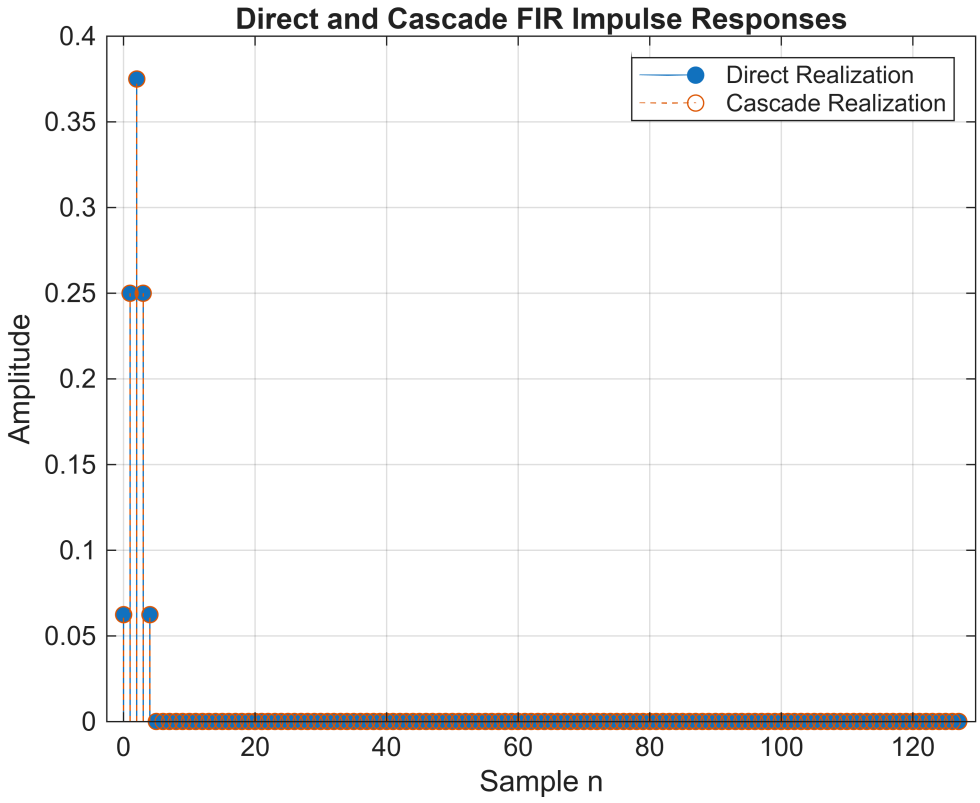
```
disp(Output_Table);
```

Parameter	Observation	V
{'Direct realization output' }	{'Direct FIR filtering' }	{'[0.0625 0.2500 0.3750
{'Cascade realization output'}	{'Cascade of first-order sections' }	{'[0.0625 0.2500 0.3750
{'Impulse response' }	{'Equal to FIR coefficient sequence' }	{'[0.0625 0.2500 0.3750
{'Phase response' }	{'Linear phase' }	{'Group delay = 2.00 sa
{'Comparison of realizations'}	{'Equivalent transfer-function behavior'}	{'Max error = 8.327e-16

```
%% 10. GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSE
figure('Name','Impulse Response Comparison');

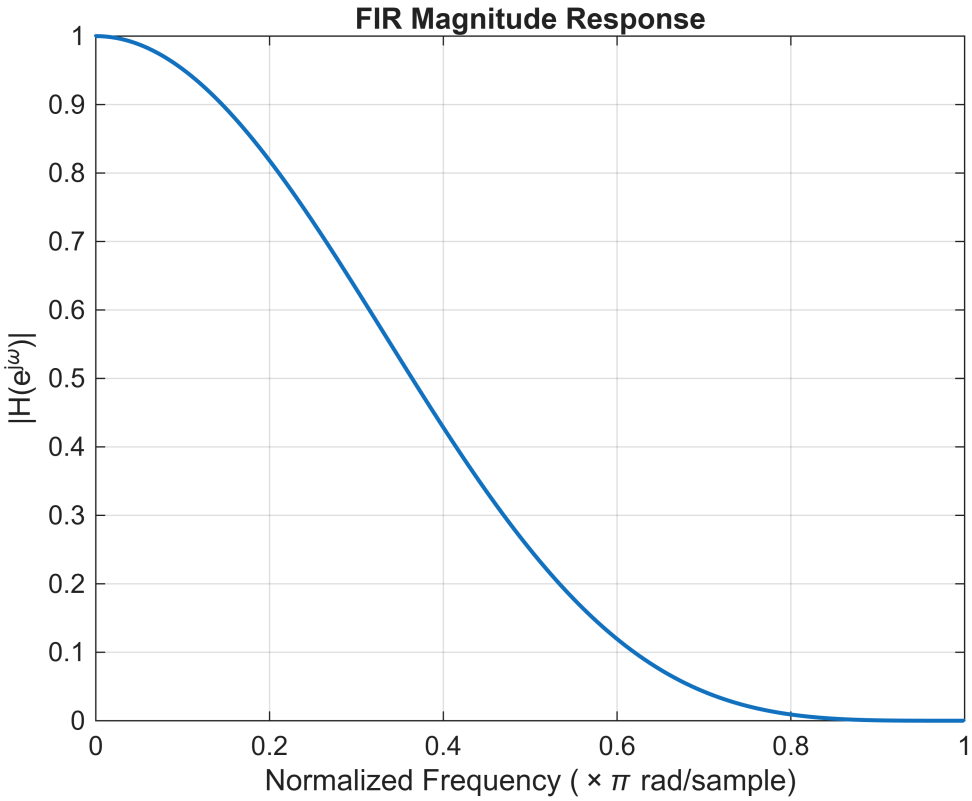
stem(n,yd,'filled');
hold on;
stem(n,yc,'--');
```

```
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct Realization','Cascade Realization');
```



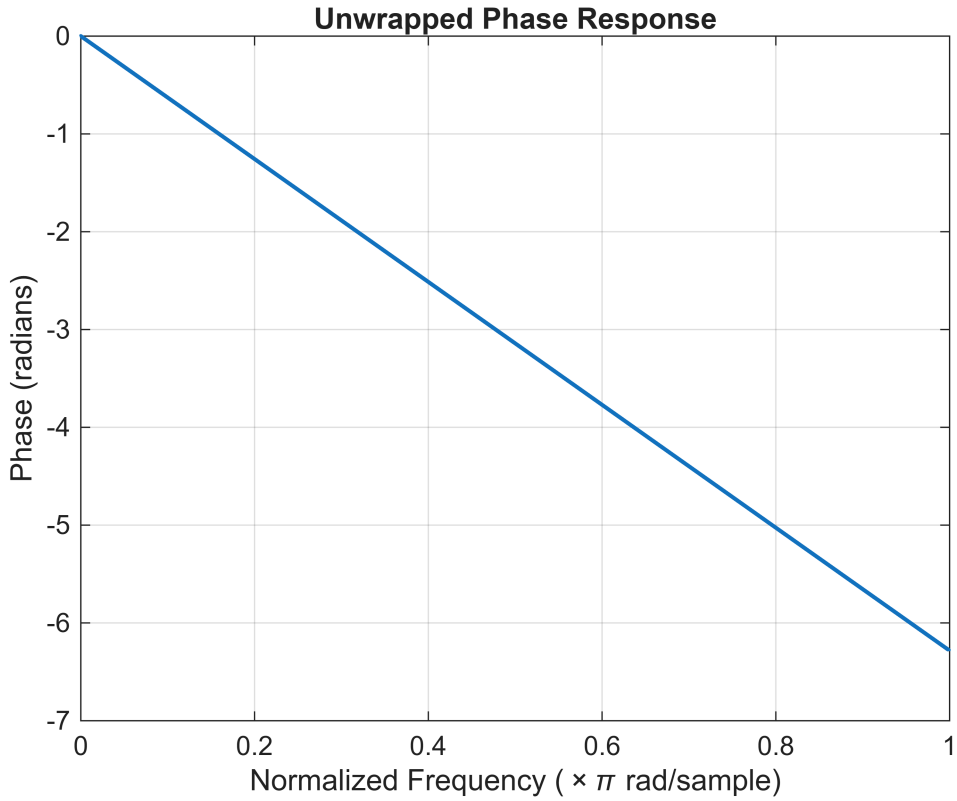

```
%% 11. GRAPH 2: MAGNITUDE RESPONSE
figure('Name','FIR Magnitude Response');

plot(w/pi,abs(H),'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');
title('FIR Magnitude Response');
```



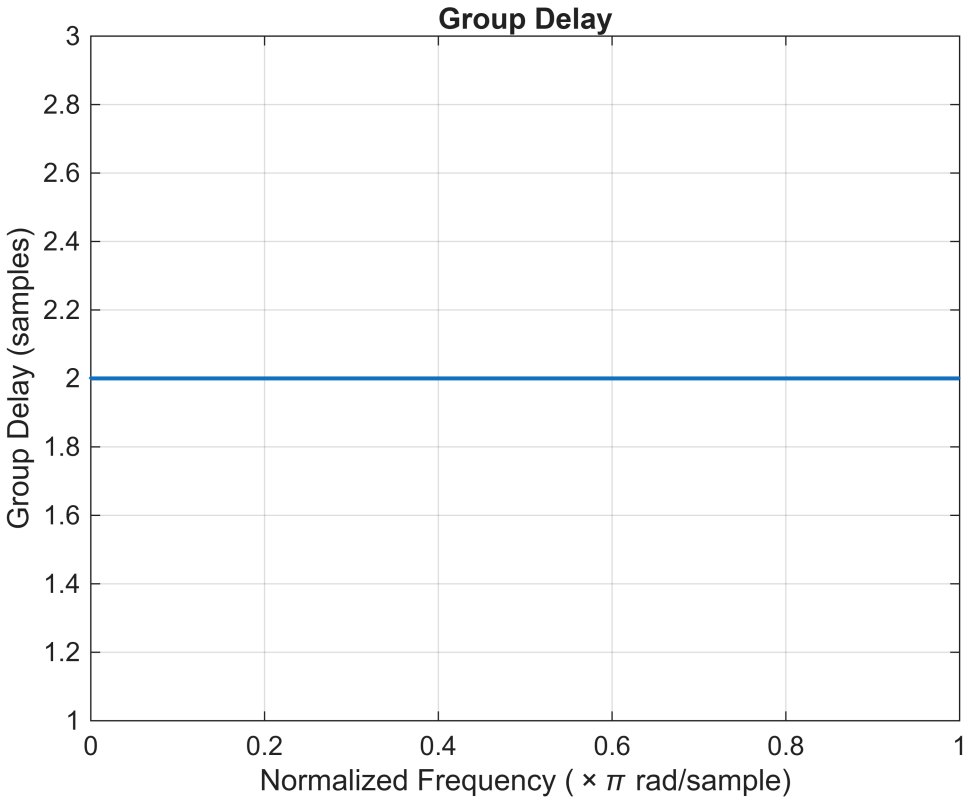

```
%% 12. GRAPH 3: PHASE RESPONSE
figure('Name','FIR Phase Response');

plot(w/pi,phase,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');
title('Unwrapped Phase Response');
```



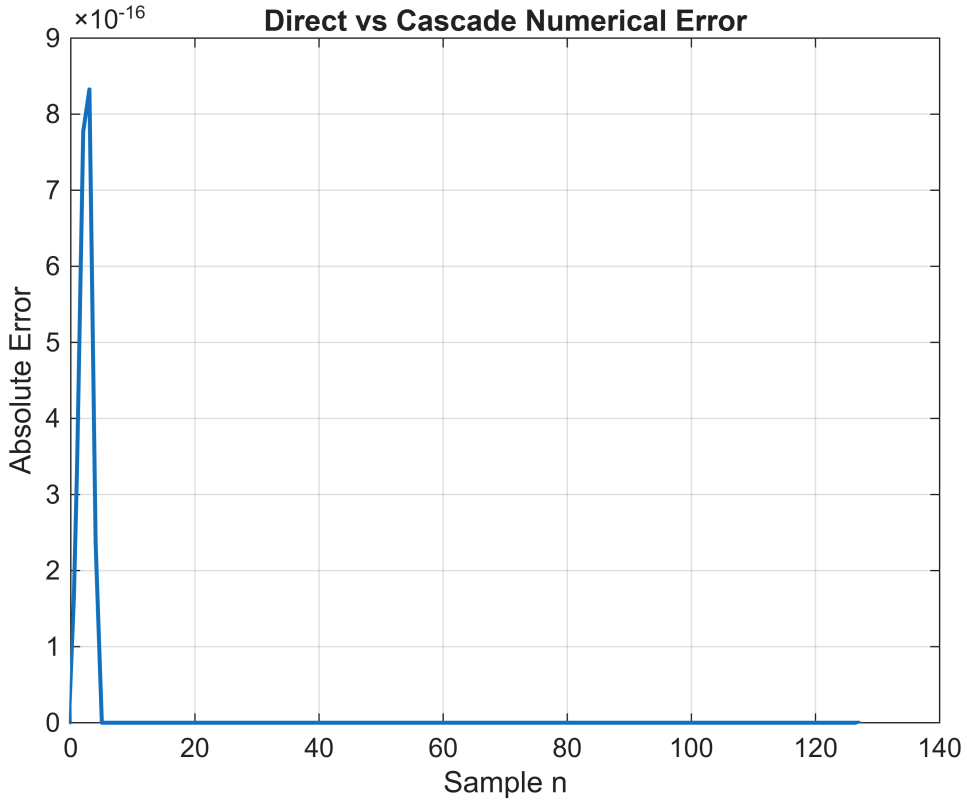

```
%% 13. GRAPH 4: GROUP DELAY
figure('Name','FIR Group Delay');

plot(w/pi,gd,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
```




```
%% 14. GRAPH 5: NUMERICAL ERROR
figure('Name','Numerical Error');

plot(n,abs(yd-yc),'LineWidth',1.5);
grid on;
xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');
```




```
%% 15. FINAL RESULT
fprintf('\n=============================================\n');
```

```
=============================================
```

```
fprintf('FINAL RESULT\n');
```

```
FINAL RESULT
```

```
fprintf('=============================================\n');
```

```
=============================================
```

```
fprintf('1. FIR filter realized using Direct structure.\n');
```

```
1. FIR filter realized using Direct structure.
```

```
fprintf('2. FIR filter realized using Cascade structure.\n');
```

```
2. FIR filter realized using Cascade structure.
```

```
fprintf('3. Coefficients are %s.\n',symmetry);
```

```
3. Coefficients are Symmetric.
```

```
fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
```

```
4. Linear-phase group delay = 2.00 samples.
```

```
fprintf('5. Maximum realization error = %.3e.\n',err);
```

```
5. Maximum realization error = 8.327e-16.
```

```
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
```

```
6. Percentage deviation = 2.220e-13 %.
```

```
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
7. Direct and Cascade responses are equivalent.
```

```

```